Ashish Gupta

Senior Software Engineer • Distributed Systems & Applied AI

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Senior Software Engineer with 9+ years of experience building distributed backend systems, event-driven platforms, and developer tooling. Currently at Tricentis in Vienna, where I build production Model Context Protocol (MCP) servers bridging LLMs with test environments, instrument high-throughput data engines with OpenTelemetry, and design real-time pub/sub pipelines with Azure Service Bus and SignalR. Strong background across C#, TypeScript, Rust, and Azure cloud services. Based in Vienna with unrestricted Austrian work authorization (Rot-Weiß-Rot-Karte Plus).

CORE COMPETENCIES

- AI & Model Context Protocols (MCP): Production MCP Servers (JSON-RPC, stdio & HTTP transports), strictly-typed tool/resource schemas, LLM agent tool-calling, NL-to-SQL synthesis, automated schema discovery, agent-facing API ergonomics (token cost, determinism, error surfacing).
- Observability & Distributed Systems: OpenTelemetry instrumentation, distributed tracing, metrics & log aggregation, agentic workflow tracing (tool calls, prompt chains, latency attribution), Event-Driven Architectures, Azure Service Bus (pub/sub), SignalR (WebSockets), gRPC.
- Cloud & Hyperscalers (Azure): Azure (Azure OpenAI, AKS, Azure Monitor, Azure Service Bus, Blob Storage, Azure Maps), Docker, Kubernetes, Helm, GitHub Actions CI/CD, Infrastructure as Code (NUKE), Snyk.
- Programming Languages & Runtimes: TypeScript, C# / .NET 8, Rust (low-latency zero-copy streaming), SQL (T-SQL, PostgreSQL), JavaScript, Java, Python.
- Developer Tooling & Web: Monaco Editor integration, React, Next.js, Angular, Nx monorepos, REST & gRPC API versioning, SDK design.
- Data Stores & Formats: SQL Server, MongoDB, Apache Parquet, Avro, Apache Hive, SQLite, 100+ GB Row-by-Row data reconciliation.

PROFESSIONAL EXPERIENCE

Tricentis — Senior Software Engineer Vienna, Austria | June 2024 — Present

Formally promoted to Senior Software Engineer in January 2026 within 19 months of tenure

- Enterprise MCP Servers: Engineered custom Model Context Protocols (MCPs) bridging LLM agents with hybrid-cloud and on-premise test environments, integrating strictly-typed tool schemas for real-time schema discovery, NL-to-SQL translation, and automated test-case generation.
- Decoupled Pub/Sub Messaging: Architected decoupled event-driven pub/sub messaging pipeline using Azure Service Bus, a dedicated Notification Service, and SignalR WebSockets, propagating state updates from autonomous MCP tool calls to client frontends with debounced live refresh.
- Observability & OpenTelemetry: Integrated .NET 8 microservices and data pipelines with OpenTelemetry and Azure Distributed Tracing, establishing deep end-to-end visibility, bottleneck profiling, and telemetry across high-volume pipelines.
- 100+ GB Data Verification Engine: Maintained, optimized, and stabilized an industry-grade Row-by-Row comparison engine reconciling 100+ GB datasets across SQL databases, Apache Hive, and columnar/binary formats (Parquet, Avro, CSV) using hybrid .NET 8 and Rust zero-copy streaming.
- Developer SQL Tooling (Monaco Editor): Developed an interactive web-based SQL playground micro-frontend using Monaco Editor with live syntax IntelliSense and low-latency result visualization.
- CI/CD Intelligence Platform: Architected GitHub Actions automated test reporting pipelines and interactive dashboard for 1,396+ nightly regression runs, adopted across 19 engineering teams for daily failure triage.

International Atomic Energy Agency (via e-Zest) — Senior Software Engineer Vienna, Austria | April 2023 — May 2024

- Mission-Critical Systems: Enhanced and maintained the SAFIRE reporting application using .NET Core, Angular, and SQL Server for international nuclear safeguards inspectors.
- Quality Control & Auditing: Led delivery of 10+ core system capabilities, reducing data validation errors by 25% under strict international security standards.

LTIMindtree — Module Lead Mumbai, India | April 2021 — April 2023

- Enterprise Compliance Platform: Engineered full-stack Remote Worker Assessment application for Deloitte (U.S.) using .NET and Angular, guaranteeing 100% regulatory compliance across 15+ international labor jurisdictions.
- Microservices Decoupling: Drove backend microservice decoupling, performance tuning, and application security hardening.

GEP Worldwide — Senior Software Engineer Mumbai, India | September 2019 — April 2021

Accelerated promotion from Software Engineer to Senior Software Engineer within 12 months (Q4 2020)

- Cloud Mapping & Telemetry: Integrated Azure Maps with Control Tower UI for real-time logistics telemetry, cutting client shipment status queries by 25%.
- UI Frameworks: Created reusable UI frameworks and interactive analytics dashboards for Control Tower and Demand Planning with NoSQL data ingestion.

TSS Consultancy — Junior Software Engineer Mumbai, India | April 2018 — September 2019

- Cross-Platform Cryptography: Built banking security modules using PKI Digital Certificates and Public/Private key cryptography, bridging security protocols between Java and .NET environments.

EDUCATION

- Bachelor of Computer Science | University of Mumbai, Mumbai, India (2012 — 2016)
  Core topics: Distributed Systems, Data Structures & Algorithms, Database Management Systems, Software Engineering.

LANGUAGES & WORK AUTHORIZATION

- Work Authorization: Rot-Weiß-Rot-Karte Plus (RWR+) — Unrestricted Austrian work permit, immediate availability with no sponsorship required.
- Languages: English (Full professional proficiency, C1/C2), German (Elementary / Advancing, A2/B1 conversational), Hindi (Native).

HONORS AND AWARDS

- Rapid Advancement at Tricentis: Formally promoted to Senior Software Engineer in January 2026 within 19 months of tenure.
- Accelerated Promotion at GEP Worldwide: Formally promoted to Senior Software Engineer within 12 months (Q4 2020).